An upper bound on the Berry–Esseen constant for independent summands

Logan Bell

6 October 2026

Abstract

Let $X_1, \dots, X_n$ be independent centered real random variables with positive total variance $V = \sum_{j=1}^n \mathbb{E}X_j^2$ and finite third absolute moments $R = \sum_{j=1}^n \mathbb{E}|X_j|^3$. We prove that the Kolmogorov distance Δ between the standardized sum $V^{-1/2}\sum_{j=1}^n X_j$ and the standard normal distribution satisfies $\Delta \le U_* L$, where $L = R/V^{3/2}$ is the Lyapunov ratio and $U_* < 0.44988$. The proof combines Prawitz's smoothing inequality with two refinements of the zero-bias characteristic-function method. First, we measure each summand's departure from a symmetric two-point distribution by its third-moment excess and aggregate the resulting single-summand zero-bias bounds by their least concave majorant. Second, we decompose the difference between the sum's characteristic function and the Gaussian characteristic function through the product of cosines $\prod_{j=1}^n \cos(t\sqrt{\mathbb{E}X_j^2/V})$, which is real and lies below $e^{-t^2/2}$ before its first zero; this allows the smoothing integral to retain the sign of the reference displacement while bounding the comparison error through a first-order differential inequality. Isolating a summand of largest variance sharpens both the product decay and the comparison forcing bound. Analytic estimates cover small and large values of L, and a rigorous interval computation covers the intermediate moment domain.

Contents

1 Introduction and main result **2**
1.1 The Berry–Esseen problem and main theorem . . . 2
1.2 Overview of the approach and open questions . . . 3
1.3 Normalization and moment parameters . . . 4

2 Bounds for characteristic-function products **5**
2.1 Cubic and trigonometric bounds . . . 5
2.2 Quadratic logarithmic correction . . . 6
2.3 Tangent lower bounds for the full logarithmic series . . . 7

3 Real and imaginary zero-bias errors **9**

4 A one-sided lower bound for the real zero-bias difference **10**

5 Quadratic and affine bounds in the third-moment excess **11**
5.1 Aggregation across summands by the least concave majorant . . . 12
5.2 Certified affine bounds in the third absolute moment . . . 13
5.3 Single-summand modulus bound from third-moment excess . . . 14

6 Isolating a summand of maximal variance **14**

7 Signed comparison with a product of cosines **15**
7.1 Bounding the forcing term 16
7.2 Sharpening the forcing bound via the largest summand 16
7.3 Joint bound on the convex weight maximum and Cauchy–Schwarz defect 18
7.4 Enclosing the cosine product and integrating the differential recurrence 18

8 Prawitz smoothing and threshold partitions **19**
8.1 Cell integration and the Gaussian correction 19
8.2 Signed smoothing over a threshold partition 20
8.3 Enclosing the signed Gaussian defect $\Gamma_{T,a}(x)$ 21

9 The certified interval calculation **21**
9.1 Analytic bounds for small and large L 21
9.2 Partition of the intermediate moment domain 22
9.3 Arithmetic conventions and verification mapping 22

A Proof of the scalar majorant for the imaginary error **23**

B Proof of the scalar bound for the cosine comparison **23**

1 Introduction and main result

1.1 The Berry–Esseen problem and main theorem

Let $X_1, \dots, X_n$ be a finite family of independent centered real random variables satisfying

$$0 < V = \sum_{j=1}^{n} \mathbb{E}X_j^2 < \infty, \qquad R = \sum_{j=1}^{n} \mathbb{E}|X_j|^3 < \infty.$$

Denote by Φ and ϕ the standard normal distribution function and density, and define the standardized sum, the Lyapunov ratio, and the Kolmogorov normal-approximation error by

$$S = \frac{\sum_{j=1}^{n} X_j}{\sqrt{V}}, \qquad L = \frac{R}{V^{3/2}}, \qquad \Delta = \sup_{x\in\mathbb{R}} |\mathbb{P}(S \le x) - \Phi(x)|. \tag{1.1}$$

The universal Berry–Esseen constant for independent summands is

$$C_{\mathrm{ind}} = \sup \frac{\Delta}{L},$$

where the supremum is taken over all $n \ge 1$ and all independent centered arrays $(X_1, \dots, X_n)$ satisfying (1.1).

In 1956, Esseen [1] proved that among identically distributed summands, the largest asymptotic value of Δ/L as $n \to \infty$ is achieved by two-point distributions with atom probabilities $(4-\sqrt{10})/2$ and $(\sqrt{10}-2)/2$, yielding the lower bound

$$C_{\mathrm{E}} = \frac{\sqrt{10}+3}{6\sqrt{2\pi}} = 0.40973218370239634299\ldots \le C_{\mathrm{ind}}; \tag{1.2}$$

see also the discussion in Shevtsova [12]. It is a longstanding open conjecture that $C_{\mathrm{ind}} = C_{\mathrm{E}}$. For unrestricted independent summands, the best previously published upper bound is $C_{\mathrm{ind}} \le 0.5583$, due to Shevtsova [11].

The main result of this paper is the following unconditional upper bound.

Theorem 1.1. *For every finite independent array satisfying* (1.1),

$$\Delta \le U_* L, \qquad C_{\mathrm{E}} \le C_{\mathrm{ind}} \le U_* < 0.44988, \\ U_* = \frac{1013039690742047}{2251799813685248}. \tag{1.3}$$

1.2 Overview of the approach and open questions

The proof of Theorem 1.1 builds on the zero-bias transformation introduced by Goldstein and Reinert [3] and the characteristic-function and smoothing methods developed by Prawitz [9, 10], Tyurin [13], Goldstein [2], and Shevtsova [12]. When S is normalized to have variance one, its characteristic function $f_S(t) = \mathbb{E}e^{\mathrm{i}tS}$ satisfies $f_S'(t) = -t f_{S^*}(t)$, where S^* has the zero-biased distribution of S. Comparing this identity with the Gaussian differential equation $(e^{-t^2/2})' = -te^{-t^2/2}$ gives

$$f_S(t) - e^{-t^2/2} = e^{-t^2/2} \int_0^t u(f_S(u) - f_{S^*}(u)) e^{u^2/2}\, du.$$

By the independent-sum replacement property of zero biasing,

$$f_S(u) - f_{S^*}(u) = \sum_{i=1}^n v_i (f_{X_i}(u) - f_{X_i^*}(u)) \prod_{j \ne i} f_{X_j}(u),$$

where $v_i = \mathbb{E}X_i^2$. In classical zero-bias arguments, one bounds the modulus of each summand difference by $|f_{X_i}(u) - f_{X_i^*}(u)| \le u\mathbb{E}|X_i|^3/(2v_i)$ and multiplies by an upper bound on $\prod_{j\ne i}|f_{X_j}(u)|$.

While this direct modulus bound is sharp for a single two-point summand as $u \to 0$, it leaves three substantial sources of overestimation at moderate frequencies and unequal variances:

(i) In the single-factor bound $|f_{X_j}(t)| \le \sqrt{1 - 2q_j(t)}$, passing to the linear exponential bound $\exp(-\sum_j q_j(t))$ discards the nonlinear terms in $-\frac{1}{2}\log(1-2q_j) = q_j + q_j^2 + \cdots$.

(ii) Bounding $|f_{X_i}(u) - f_{X_i^*}(u)|$ by $u\mathbb{E}|X_i|^3/(2v_i)$ ignores the fact that when $\mathbb{E}|X_i|^3$ is close to its minimum $v_i^{3/2}$, the law of $X_i/\sqrt{v_i}$ must be close to the symmetric Rademacher distribution on $\{-1, 1\}$, for which $(f_W(x) - f_{W^*}(x))/x = (\cos x - \operatorname{sinc} x)/x$ has a strictly smaller coefficient than $1/2$ for $x > 0$.

(iii) Taking the absolute value $|f_S(t) - e^{-t^2/2}|$ inside Prawitz's smoothing integral discards the sign of the characteristic-function displacement when the summands are nearly symmetric two-point variables.

Our proof addresses each of these three mechanisms in turn:

- **Logarithmic product bounds (Section 2):** We retain both the quadratic term $\sum_j q_j^2$ and certified tangent-line lower bounds for the full series $-\frac{1}{2}\log(1-2q_j) - q_j$, expressed in terms of L and the variance power sum $\tau = \sum_{j=1}^n v_j^{3/2}$.

- **Third-moment excess and concave majorants (Sections 3–5):** For a mean-zero, variance-one variable W with $\rho = \mathbb{E}|W|^3$, the identity $\mathbb{E}[(|W|-1)^2(|W|+1/2)] = \rho - 1 \geq 0$ controls how far $f_W(x)$ and $f_{W^*}(x)$ can deviate from $\cos x$ and $\operatorname{sinc} x$. We derive a two-sided interval for $\Re(f_W - f_{W^*})$, a quadratic bound in $e = \rho - 1$ for $|\Im(f_W - f_{W^*})|^2$, and certified affine modulus and directional bounds. Taking the least concave majorant in e aggregates these single-summand bounds across arrays with arbitrary variances.
- **Isolation of a largest-variance summand and signed cosine comparison (Sections 6–7):** Writing $d = \max_j v_j$ and b for the third absolute moment of a summand achieving variance d, we separate that summand from the remaining array in both the zero-bias product and the reference comparison. Following the symmetric Bernoulli approximation studied by Mattner and Shevtsova [6, Theorem 1.2 and Example 1.4], we compare $f_S(t)$ with the cosine product $C_v(t) = \prod_{j=1}^n \cos(t\sqrt{v_j})$ via a first-order differential equation. Because $C_v(t) - e^{-t^2/2} \leq 0$ is real prior to the first zero of the cosines, its negative sign is preserved in the threshold-localized signed Prawitz smoothing inequality.
- **Smoothing and certified interval cover (Sections 8–9):** Analytic estimates bound Δ/L for $L \leq L_- \approx 0.0014$ (using Corollary 4.18 of Shevtsova [12]) and for $L \geq L_+ \approx 1.21$ (using Cantelli's inequality). Over $L \in [L_-, L_+]$, a rigorous interval computation partitions the feasible region of (L, d, b, τ) into closed boxes and verifies $\Delta/L \leq U_* < 0.44988$ on every box.

Closing the remaining gap between $C_{\mathrm{E}} \approx 0.409732$ and $U_* < 0.44988$ on the intermediate range of L remains open. In Fourier-smoothing arguments, the principal remaining source of loss is the triangle inequality applied across summands in the zero-bias and cosine-reference forcing terms, which discards phase cancellation between summands of different skewnesses and lattice spans. Companion papers in this collection establish the sharp bound $\Delta \leq C_{\mathrm{E}} L$ for $L \leq \exp(-\exp(97))$, for arrays of at most two arbitrary summands ($C_2 < 49/120 < C_{\mathrm{E}}$), and for several classes of two-point arrays.

1.3 Normalization and moment parameters

After replacing each X_j by $X_j/\sqrt{V}$ and omitting any summands of variance zero, we have $S = \sum_{j=1}^n X_j$ with total variance one. Write

$$v_j = \mathbb{E}X_j^2, \qquad b_j = \mathbb{E}|X_j|^3, \qquad \tau = \sum_{j=1}^n v_j^{3/2}, \qquad \sum_{j=1}^n v_j = 1, \qquad \sum_{j=1}^n b_j = L.$$

By Lyapunov's inequality, each difference $b_j - v_j^{3/2}$ is nonnegative, vanishing if and only if X_j is a symmetric two-point variable on $\{-\sqrt{v_j}, \sqrt{v_j}\}$, and $\sum_{j=1}^n (b_j - v_j^{3/2}) = L - \tau \geq 0$.

Relabel the summands so that $v_1 = \max_{1\leq j\leq n} v_j$, and set $d = v_1$ and $b = b_1$. The four moment parameters (L, d, b, τ) satisfy the necessary constraints

$$b \geq d^{3/2}, \qquad d^{3/2} \leq \tau \leq \min\{L - b + d^{3/2}, \sqrt{d}\}, \qquad \tau \leq L. \tag{1.4}$$

Indeed, $b_j \geq v_j^{3/2}$ for $j \geq 2$ implies $\tau - d^{3/2} \leq L - b$, while $v_j \leq d$ implies $\tau \leq \sqrt{d}\sum_{j=1}^n v_j = \sqrt{d}$; in particular, $\tau^2 \leq d \leq \tau^{2/3}$. For the subarray of remaining summands $j \in \{2, \ldots, n\}$, we define the remainder parameters

$$w = 1 - d, \qquad B_R = L - b, \qquad \tau_R = \tau - d^{3/2}, \qquad a_R = \min\{d, w, B_R^{2/3}, \tau_R^{2/3}\}. \tag{1.5}$$

Every remaining summand $j \geq 2$ satisfies $v_j \leq d$, $v_j \leq w$, and $v_j^{3/2} \leq \min\{b_j, \tau_R\} \leq \min\{B_R, \tau_R\}$, so that $v_j \leq a_R$. Every finite independent array realizes a parameter tuple (L, d, b, τ) in the domain defined by (1.4)–(1.5), and an upper bound valid across this domain bounds Δ/L for every array.

2 Bounds for characteristic-function products

For a real random variable X, write $f_X(t) = \mathbb{E}e^{itX}$ for its characteristic function, and set $f_j = f_{X_j}$. Because $f_X(-t) = \overline{f_X(t)}$, it suffices to bound $|f_X(t)|$ for $t \geq 0$. Given a nonempty subset of indices $J \subseteq \{1, \dots, n\}$, put

$$V_0 = \sum_{j \in J} v_j, \qquad B = \sum_{j \in J} b_j, \qquad \tau_0 = \sum_{j \in J} v_j^{3/2},$$

and let $a > 0$ be an upper bound on $\max_{j \in J} v_j$. We require upper bounds both for the full product $\prod_{j \in J} |f_j(t)|$ and for the deleted products $\prod_{j \in J \setminus \{i\}} |f_j(t)|$ obtained by omitting any single index $i \in J$ (with empty products equal to 1). Deleted products enter directly through the zero-bias identity (6.1), which implies

$$|f_S(t) - f_{S^*}(t)| \leq \sum_{i=1}^{n} v_i |f_{X_i}(t) - f_{X_i^*}(t)| \prod_{j \neq i} |f_j(t)|.$$

2.1 Cubic and trigonometric bounds

Define the constant

$$\kappa_* = \sup_{x > 0} \frac{\cos x - 1 + x^2/2}{x^3}, \qquad \kappa = \frac{49581}{500000} = 0.099162, \tag{2.1}$$

and let $\theta \in (\pi, 2\pi)$ denote the unique maximizer in (2.1). For $V_0, E_0 > 0$, $t \geq 0$, and $u = E_0 t / V_0$, set

$$\psi(t; V_0, E_0) = \begin{cases} V_0 t^2/2 - \kappa_* E_0 t^3, & u < \theta, \\ V_0^3 E_0^{-2}(1 - \cos u), & \theta \leq u \leq 2\pi, \\ 0, & u > 2\pi. \end{cases} \tag{2.2}$$

The characteristic-function inequalities of Prawitz [10], in the form given by Tyurin [13, Lemma 1], yield for any centered random variable X of variance $v > 0$ and third absolute moment $b \geq v^{3/2}$,

$$\Big| \prod_{j \in J} f_{X_j}(t) \Big| \leq \exp\{-\psi(t; V_0, B + \tau_0)\}, \tag{2.3}$$

$$|f_X(t)| \leq A(t; v, b) := \sqrt{1 - 2\psi(t; v, b + v^{3/2})}. \tag{2.4}$$

Let us record the monotonicity properties of ψ used to evaluate (2.3)–(2.4) on parameter intervals. For fixed (V_0, E_0), $\psi(t; V_0, E_0)$ increases with t on $[0, V_0/(3\kappa_* E_0)]$, decreases on $[V_0/(3\kappa_* E_0), 2\pi V_0/E_0]$, and vanishes for $t \geq 2\pi V_0/E_0$: indeed, $\partial_t \psi = t(V_0 - 3\kappa_* E_0 t)$ on the cubic branch and $\partial_t \psi = (V_0^2/E_0) \sin u \leq 0$ on the cosine branch, with matching values at $u = \theta$ by the stationarity condition for θ. Consequently, the upper bounds in (2.3) and (2.4) attain their maxima over any closed frequency interval at one of its endpoints. With respect to the parameters

(V_0, E_0), we have on the cosine branch

$$\partial_{V_0}\psi = \frac{V_0^2}{E_0^2}\{3(1-\cos u) - u\sin u\} \geq 0,$$
$$\partial_{E_0}\psi = \frac{V_0^3}{E_0^3}\{u\sin u - 2(1-\cos u)\} \leq 0,$$

and the same signs hold on the cubic branch. Thus $\psi(t; V_0, E_0)$ is nondecreasing in V_0 and nonincreasing in E_0.

To verify that the supremum in (2.1) is uniquely attained and bounded by κ, set $r(x) = (\cos x - 1 + x^2/2)/x^3$. Differentiation gives

$$r'(x) = -\frac{h(x)}{2x^4}, \quad h(x) = x^2 + 2x\sin x + 6(\cos x - 1), \quad h'(x) = 4\cos(x/2)\{x\cos(x/2) - 2\sin(x/2)\}.$$

Because $x\cos(x/2) - 2\sin(x/2)$ is strictly negative on $(0, 2\pi)$, h decreases on $(0, \pi)$ and increases on $(\pi, 2\pi)$, with $h(\pi) = \pi^2 - 12 < 0$ and $h(2\pi) = 4\pi^2 > 0$. Thus h has a unique zero $\theta \in (\pi, 2\pi)$. For $x \geq 2\pi$, $r(x) \leq 1/(4\pi) < 3/32 \leq r(4)$, so θ is the unique global maximizer of r on $(0, \infty)$. The scalar checker isolates $\theta \in (3.99, 4)$ and encloses $\kappa_* < 0.099161913515 < \kappa$ using Arb ball arithmetic [4]. In interval evaluations, the cubic lower bound $V_0 t^2/2 - \kappa E_0 t^3 \leq \psi(t; V_0, E_0)$ is valid at all frequencies, and the cosine branch is used when the entire interval of u lies in $[4, 2\pi] \subset [\theta, 2\pi]$.

2.2 Quadratic logarithmic correction

For each $j \in J$, define

$$h_j = \frac{v_j t^2}{2} - \kappa(b_j + v_j^{3/2})t^3, \qquad q_j = \max(h_j, 0), \qquad H = \frac{V_0 t^2}{2} - \kappa(B + \tau_0)t^3.$$

Since $b_j \geq v_j^{3/2}$ and $v_j \leq a$, we have

$$0 \leq q_j \leq m(t, a) := \max_{0 \leq x \leq t\sqrt{a}} (x^2/2 - 2\kappa x^3) \leq \frac{1}{216\kappa^2} < \frac{1}{2}, \tag{2.5}$$

where the maximum in $m(t, a)$ occurs at $x = \min(t\sqrt{a}, 1/(6\kappa))$. From (2.4) we have $|f_j(t)| \leq \sqrt{1 - 2q_j}$. Applying the inequality $-\frac{1}{2}\log(1-2q) \geq q + q^2$ for $0 \leq q < 1/2$ and noting that $\sum_{j\in J} q_j \geq \sum_{j\in J} h_j = H$, we obtain

$$\prod_{j\in J} |f_j(t)| \leq \exp\left\{-H - \sum_{j\in J} q_j^2\right\}. \tag{2.6}$$

Lemma 2.1. *Set $A_0 = t^2/2 - 2\kappa\sqrt{a}\, t^3$. When $A_0 \geq 0$,*

$$\sum_{j\in J} q_j^2 \geq Q := \left(\max\left\{0, A_0 \frac{\tau_0}{\sqrt{V_0}} - \kappa t^3(B - \tau_0)\right\}\right)^2. \tag{2.7}$$

When $A_0 < 0$, the bound (2.7) holds with $Q = 0$.

Proof. Since $v_j \leq a$ and $q_j \geq h_j$, each summand satisfies

$$q_j + \kappa t^3(b_j - v_j^{3/2}) \geq v_j(t^2/2 - 2\kappa\sqrt{v_j}\, t^3) \geq A_0 v_j.$$

When $A_0 \geq 0$, the triangle inequality in $\ell_2(J)$ gives

$$\|(q_j)\|_2 \geq A_0\|(v_j)\|_2 - \kappa t^3(B - \tau_0).$$

By Cauchy–Schwarz, $\tau_0 = \sum_{j\in J} v_j^{3/2} \leq \sqrt{V_0}\|(v_j)\|_2$. Substituting this lower bound for $\|(v_j)\|_2$, taking the positive part, and squaring yields (2.7). □

When a single index $i \in J$ is omitted, the linear sum satisfies $\sum_{j\neq i} q_j \geq H - h_i \geq H - m(t,a)$, while the quadratic sum satisfies $\sum_{j\neq i} q_j^2 \geq \max(0, Q - m(t,a)^2)$. We therefore obtain the full and deleted product bounds

$$\begin{aligned} F_0(t; V_0, B, a, \tau_0) &= \min\{1, e^{-H-Q}\}, \\ D_0(t; V_0, B, a, \tau_0) &= \min\{1, e^{-H+m(t,a)-\max(0,Q-m(t,a)^2)}\}, \end{aligned} \tag{2.8}$$

setting $F_0 = D_0 = 1$ for the empty index set.

To maximize the bounds (2.8) as τ_0 varies over a closed interval $[\ell, h]$, suppose $A_0 \geq 0$ and $t > 0$, and put $k = \kappa t^3$, $a_0 = A_0/\sqrt{V_0} + k$, and $c_0 = kB$. Up to terms independent of τ_0, the negative of the exponent in either bound is

$$k\tau_0 - \max\{0, (a_0\tau_0 - c_0)_+^2 - m_0^2\}, \qquad z_+ = \max(z, 0),$$

with $m_0 = 0$ for F_0 and $m_0 = m(t,a)$ for D_0. This function of τ_0 has derivative $k > 0$ for $a_0\tau_0 - c_0 \leq m_0$ and derivative $k - 2a_0(a_0\tau_0 - c_0)$ for $a_0\tau_0 - c_0 > m_0$, so its maximum on $[\ell, h]$ is attained at

$$\min\{h, \max\{\ell, \tau_*\}\}, \qquad \tau_* = \frac{c_0 + \max\{m_0, k/(2a_0)\}}{a_0}.$$

When $A_0 < 0$, $Q = 0$ and the expression increases linearly with τ_0.

2.3 Tangent lower bounds for the full logarithmic series

To capture the higher-order terms in $-\frac{1}{2}\log(1-2q)$ beyond the quadratic term, put

$$g(q) = -\tfrac{1}{2}\log(1-2q) - q = \sum_{p=2}^{\infty} \frac{2^{p-1}}{p} q^p, \qquad 0 \leq q < 1/2.$$

With $k = \kappa t^3$, $\epsilon_j = b_j - v_j^{3/2} \geq 0$, and $r_j = v_j t^2/2 - 2kv_j^{3/2}$, we have $q_j = (r_j - k\epsilon_j)_+ \in [0, m(t,a)]$. For any slope parameter $\lambda \geq 0$ and $0 \leq r < 1/2$, define the infimal convolution

$$\varphi_\lambda(r) = \inf_{0\leq q\leq r}\{g(q) + \lambda(r-q)\} = \begin{cases} g(r), & r \leq q_\lambda, \\ \lambda r - c_\lambda, & r \geq q_\lambda, \end{cases} \quad q_\lambda = \frac{\lambda}{2(1+\lambda)}, \quad c_\lambda = \frac{\lambda - \log(1+\lambda)}{2}, \tag{2.9}$$

which follows directly from $\partial_q\{g(q) + \lambda(r-q)\} = 2q/(1-2q) - \lambda$.

Lemma 2.2 (Logarithmic correction via cubic lower bounds)**.** *Set $A = 1/(32\kappa^2)$ and $r(z) = Az^2(1-z)$. Suppose $\alpha, \beta \geq 0$ and $0 < M \leq 1$ satisfy*

$$\varphi_\lambda(r(z)) \geq z^2(\alpha z - \beta) \qquad (0 \leq z \leq M). \tag{2.10}$$

If $4\kappa t\sqrt{a} \leq M$, define

$$S_\lambda = \frac{V_0 t^2}{2} - (1+\lambda)kB - 16\kappa^2 t^2 \beta V_0 + \{64\kappa^3 t^3 \alpha + (\lambda - 1)k\}\tau_0, \tag{2.11}$$

$$P(v) = 64\kappa^3 t^3 \alpha v^{3/2} - 16\kappa^2 t^2 \beta v, \qquad \delta = \max\{0, P(a)\}. \tag{2.12}$$

Then for every index $i \in J$,

$$\prod_{j\in J} |f_j(t)| \le \min\{1, e^{-S_\lambda}\}, \qquad \prod_{j\in J\setminus\{i\}} |f_j(t)| \le \min\{1, e^{-S_\lambda + m(t,a)+\delta}\}.$$

Proof. The condition $4\kappa t\sqrt{a} \le M \le 1$ ensures $z_j = 4\kappa t\sqrt{v_j} \in [0, M]$ and $r_j = r(z_j) \ge 0$. Since $0 \le q_j \le r_j$ and $r_j - q_j \le k\epsilon_j$, the definition (2.9) and (2.10) give

$$\sum_{j\in J} g(q_j) \ge \sum_{j\in J} \varphi_\lambda(r_j) - \lambda k(B - \tau_0) \ge 64\kappa^3 t^3 \alpha\tau_0 - 16\kappa^2 t^2 \beta V_0 - \lambda k(B - \tau_0).$$

Adding $\sum_{j\in J} q_j \ge H$ yields $\sum_{j\in J}[q_j + g(q_j)] \ge S_\lambda$, which proves the full-product bound. Deleting index i removes $q_i \le m(t, a)$ from the linear sum and $P(v_i) \le \max\{P(0), P(a)\} = \delta$ (by convexity of P on $[0, a]$) from the support sum, while bounding the remaining moment excess by $B - \tau_0$ preserves the inequality. □

The following criterion reduces verification of (2.10) on $[0, M]$ to checking its right endpoint $z = M$.

Lemma 2.3 (Endpoint criterion for tangent lower bounds)**.** *Let $\psi(z) = g(r(z))/z^2$ (extended by continuity at $z = 0$) and $D(z) = z^4\psi''(z)$. If $0 < u < 2/3$ satisfies $r(u) \le q_\lambda$ and $D(u) \ge 0$, then $\alpha = \psi'(u)$ and $\beta = u\psi'(u) - \psi(u)$ are nonnegative, and for any $M \in [u, 1]$, the inequality (2.10) holds on $[0, M]$ if and only if*

$$\frac{\varphi_\lambda(r(M))}{M^2} \ge \alpha M - \beta.$$

Proof. Direct differentiation gives

$$D(z) = g''(r)\{Az^2(2 - 3z)\}^2 - 6rg'(r) + 6g(r),$$

$$D'(z) = \frac{4A^2z^3(2 - 3z)}{(1 - 2r)^3}\, B(z), \qquad B(z) = 1 - 6z + 2Az^2(3 - 5z + 3z^2).$$

Since $\max_{[0,1]} r = 4A/27 < 1/2$, we have $A < 27/8$. On $[0, 2/3]$,

$$B'(z) = -6 + 6Az(2 - 5z + 4z^2) < 0,$$

because $8/27 - z(2 - 5z + 4z^2) = (2/3 - z)(4z^2 - 7z/3 + 4/9) \ge 0$. As $B(0) = 1$ and $B(2/3) = -3 + 8A/9 < 0$, $D(z)$ first increases from $D(0) = 0$ and then decreases to $D(2/3) = 6\{g(r) - rg'(r)\} < 0$, so D has a unique zero in $(0, 2/3)$. Thus ψ is convex from 0 up to that inflection point (which lies to the right of u since $D(u) \ge 0$), concave on the remainder of $(0, 2/3)$, and decreasing on $[2/3, 1]$. On the linear branch $r \ge q_\lambda$,

$$\left(\frac{\varphi_\lambda(r(z))}{z^2}\right)'' = \left(\lambda A(1 - z) - \frac{c_\lambda}{z^2}\right)'' = -\frac{6c_\lambda}{z^4} \le 0.$$

Since $r(z)$ strictly increases on $[0, 2/3]$ and $\varphi_\lambda(r(z))/z^2$ is C^1 across the transition $r(z) = q_\lambda$, on $[0, 2/3]$ the derivative of $\varphi_\lambda(r(z))/z^2$ increases on $[0, z_0]$ (where $z_0 \ge u$ is the smaller of the inflection point of ψ and the transition point $r(z) = q_\lambda$ on $[0, 2/3]$, or equals the inflection point of ψ when $q_\lambda > 4A/27 = \max_{[0,1]} r$) and decreases on $[z_0, 2/3]$, while on $[2/3, 1]$ it is nonpositive. Consequently, the difference $\varphi_\lambda(r(z))/z^2 - (\alpha z - \beta)$ decreases to zero at $z = u$, is nondecreasing on $[u, z_0]$, and is concave on $[z_0, 2/3]$ and nonincreasing on $[2/3, 1]$; its minimum on $[0, M]$ is therefore attained at $z = u$ or $z = M$. Finally, convexity on $[0, u]$ and $\psi(0) = \psi'(0) = 0$ imply $\alpha, \beta \ge 0$. □

The certificate `certificates/scalar/direct-log.json` stores 2,006 tangent lower bounds across $\lambda \in \{3/4, 1, 3/2, 2, 3\}$, each verified using Lemma 2.3 in 192-bit Arb arithmetic. On any parameter box, S_λ is affine in τ_0 and attains its minimum at an endpoint of the τ_0 interval (when $\lambda < 1$, we bound the nonnegative quantities $B + \tau_0$ and $B - \tau_0$ in $-k(B+\tau_0) - \lambda k(B-\tau_0)$ before expanding). We denote by $\mathcal{F}_0$ and $\mathcal{D}_0$ the pointwise minima of the quadratic bounds (2.8) and all applicable tangent bounds from Lemma 2.2 (setting $\mathcal{F}_0 = \mathcal{D}_0 = 1$ for the empty product).

3 Real and imaginary zero-bias errors

For a centered random variable X of variance $v > 0$, its zero-biased distribution X^* is characterized by

$$v\mathbb{E}g'(X^*) = \mathbb{E}[Xg(X)] \tag{3.1}$$

for all absolutely continuous functions g with $\mathbb{E}|Xg(X)| < \infty$ [3]. We write $f_j^* = f_{X_j^*}$. By the sharp L^1-Wasserstein bound $W_1(X, X^*) \le \mathbb{E}|X|^3/(2v)$ proved independently by Goldstein [2, Lemma 1.1] and Tyurin [13, Theorem 3], the Lipschitz function $x \mapsto e^{itx}$ satisfies

$$|f_X(t) - f_{X^*}(t)| \le |t|\frac{\mathbb{E}|X|^3}{2v}. \tag{3.2}$$

To refine (3.2) when $\mathbb{E}|X|^3$ is close to $v^{3/2}$, consider a standardized random variable W with $\mathbb{E}W = 0$, $\mathbb{E}W^2 = 1$, $Y = |W|$, $\rho = \mathbb{E}Y^3$, and third-moment excess $e = \rho - 1 \ge 0$. Because $\mathbb{E}Y^2 = 1$, the cubic polynomial $h(y) = (y-1)^2(y+1/2) = y^3 - \frac{3}{2}y^2 + \frac{1}{2}$ satisfies the exact identity

$$\mathbb{E}[(Y-1)^2(Y+1/2)] = e. \tag{3.3}$$

In particular, $e = 0$ if and only if $Y = 1$ almost surely, which since $\mathbb{E}W = 0$ means that W has the symmetric Rademacher distribution $\frac{1}{2}\delta_{-1} + \frac{1}{2}\delta_1$, with $f_W(x) = \cos x$ and $f_{W^*}(x) = \operatorname{sinc} x = (\sin x)/x$.

Lemma 3.1. *For every $x \in \mathbb{R}$,*

$$|\Re f_W(x) - \cos x| \le \frac{e|x|^3}{6}, \qquad |\Re f_{W^*}(x) - \operatorname{sinc} x| \le \frac{e|x|}{2}. \tag{3.4}$$

Proof. For any even function f with Lipschitz second derivative, the remainder after subtracting its quadratic Taylor approximation in y^2 at $y = 1$,

$$R_f(y) = f(y) - f(1) - \frac{f'(1)}{2}(y^2 - 1), \qquad y \ge 0,$$

admits the integral representation $R_f(y) = \int_0^{\max(1,y)} K_y(u) f'''(u)\, du$, where the Peano kernel $K_y(u)$ is nonnegative on $[0, \max(1,y)]$ (specifically, for $y \le 1$, $K_y(u) = u(y-1)^2/2$ on $[0,y]$ and $(1-u)(u-y^2)/2$ on $[y,1]$; for $y \ge 1$, $K_y(u) = u(y-1)^2/2$ on $[0,1]$ and $(y-u)^2/2$ on $[1,y]$) and has total integral $\int_0^{\max(1,y)} K_y(u)\, du = (y-1)^2(y+1/2)/6$. Applying this identity to $f(y) = \cos(xy)$, where $\|f'''\|_\infty \le |x|^3$ and $\mathbb{E}(Y^2 - 1) = 0$, and taking expectations via (3.3) yields the first bound in (3.4).

For the second bound, let $W^\square$ have the size-biased distribution defined by $\mathbb{E}g(W^\square) = \mathbb{E}[W^2 g(W)]$, so that $W^* \overset{d}{=} UW^\square$ with $U \sim \mathrm{Unif}[0,1]$ independent of $W^\square$. For every $z \ge 0$,

$$\mathbb{P}(|W^*| \le z) = \mathbb{E}[Y \min(z, Y)] \le \min(z, 1) = \mathbb{P}(U \le z),$$

so $|W^*|$ stochastically dominates U. Their L^1-Wasserstein distance is therefore equal to the difference of their means: $W_1(|W^*|, U) = \mathbb{E}|W^*| - \mathbb{E}U = \rho/2 - 1/2 = e/2$. Since $y \mapsto \cos(xy)$ is $|x|$-Lipschitz and $\Re f_{W^*}(x) = \mathbb{E}\cos(x|W^*|)$ while $\operatorname{sinc} x = \mathbb{E}\cos(xU)$, the second inequality in (3.4) follows. □

Lemma 3.2. *For every $x \in \mathbb{R}$,*

$$|\Im f_W(x)| \le \frac{|x|^3}{6}\sqrt{e(\rho + 5/3)}. \tag{3.5}$$

Proof. By symmetry we may assume $x > 0$. Set $A(y) = x\operatorname{sinc}(xy) = (\sin(xy))/y$. Since $\mathbb{E}W = 0$, we have $\Im f_W(x) = \mathbb{E}\sin(xW) = \mathbb{E}[W(A(Y) - A(1))]$. Because $|\operatorname{sinc}'(z)| \le |z|/3$, we have $|A'(y)| \le x^3 y/3$, and integrating from 1 to y gives $|A(y) - A(1)| \le (x^3/6)|y^2 - 1|$. Applying the Cauchy–Schwarz inequality and (3.3) yields

$$\mathbb{E}[Y|Y^2 - 1|] \le \sqrt{\mathbb{E}[(Y-1)^2(Y+1/2)]\,\mathbb{E}\left[\frac{Y^2(Y+1)^2}{Y+1/2}\right]} \le \sqrt{e(\rho + 5/3)},$$

where the last step uses the algebraic identity and inequality

$$\frac{y^2(y+1)^2}{y+1/2} = y^3 + \frac{3}{2}y^2 + \frac{y^2}{4(y+1/2)} \le y^3 + \frac{3}{2}y^2 + \frac{2+4y^2}{36} = y^3 + \frac{29}{18}y^2 + \frac{1}{18},$$

which holds for all $y \ge 0$ since $(2+4y^2)/9 - y^2/(y+1/2) = (y-1)^2(4y+1)/[9(y+1/2)] \ge 0$. □

4 A one-sided lower bound for the real zero-bias difference

While Lemma 3.1 bounds $|\Re(f_W(x) - f_{W^*}(x)) - (\cos x - \operatorname{sinc} x)|$ from both sides by $(x/2 + x^3/6)e$, the lower side of this interval can be improved on $0 \le x \le 2$: the Rademacher law actually minimizes $\Re(f_W(x) - f_{W^*}(x))$ among all centered unit-variance distributions.

Lemma 4.1. *For $0 < x \le 2$, set*

$$F_x(y) = \cos(xy) - y\frac{\sin(xy)}{x}, \qquad A_x = \cos x + (1 + x^2)\operatorname{sinc} x.$$

Then $F_x(y) \ge F_x(1) - \frac{1}{2}A_x(y^2 - 1)$ for all $y \ge 0$. Consequently,

$$\Re(f_W(x) - f_{W^*}(x)) \ge \cos x - \operatorname{sinc} x. \tag{4.1}$$

Proof. Put $h_x(z) = \cos z + (1 + x^2)\operatorname{sinc} z$, so that $A_x = h_x(x)$. With $z = xy$, the derivative with respect to y of $x^2\{F_x(y) - F_x(1) + \frac{1}{2}A_x(y^2 - 1)\}$ is $z[A_x - h_x(z)]$.

The function $h_x(z)$ is strictly decreasing on $(0, \pi)$, is at most 1 on $[\pi, 2\pi]$, and is at most $1 + 5/(2\pi) < 11/6$ for $z \ge 2\pi$. On the other hand, alternating Taylor series bounds give, for $0 < x \le 2$,

$$A_x \ge p_0(x^2), \qquad p_0(u) = 2 + \frac{u}{3} - \frac{7u^2}{60} + \frac{17u^3}{2520} - \frac{u^4}{5040}.$$

On $[0, 4]$, $p_0''(u) = (-98 + 17u - u^2)/420 < 0$, so $p_0(u) \ge \min(p_0(0), p_0(4)) = 194/105 > 11/6$. Thus $A_x - h_x(z)$ is negative for $0 < z < x$ and positive for $z > x$, so $F_x(y) - F_x(1) + \frac{1}{2}A_x(y^2 - 1)$ attains its global minimum on $[0, \infty)$ at $y = 1$, where it equals zero. Taking expectations under $Y = |W|$ and using $\mathbb{E}W^* = UW^{\square}$ gives $\mathbb{E}F_x(Y) = \Re(f_W(x) - f_{W^*}(x))$, proving (4.1). □

Combining Lemmas 3.1 and 4.1, we see that for $0 \le x \le 2$, with $g(x) = \operatorname{sinc} x - \cos x \ge 0$ and $c(x) = x/2 + x^3/6$,

$$-g(x) \le \Re(f_W(x) - f_{W^*}(x)) \le -g(x) + c(x)e. \tag{4.2}$$

By the alternating Taylor series of $\operatorname{sinc} x - \cos x$, we have the increasing upper bound

$$\frac{2g(x)}{x} \le \alpha(x) := \begin{cases} 2x/3 - x^3/15 + x^5/420, & 0 \le x \le 2, \\ 2x/3, & x > 2, \end{cases} \tag{4.3}$$

and we set $q(x) = 2c(x)/x = 1 + x^2/3$.

5 Quadratic and affine bounds in the third-moment excess

We next establish a quadratic bound in $e = \rho - 1$ for $|\Im(f_W(x) - f_{W^*}(x))|^2$ and combine it with (4.2) via a least concave majorant.

Lemma 5.1 (Quadratic imaginary-error bound)**.** *For $0 < x \le 2$,*

$$|\Im(f_W(x) - f_{W^*}(x))|^2 \le e \left\{ \frac{2}{3}(x\cos x - 2\sin x)^2 + \frac{x^2 e}{4} \right\}. \tag{5.1}$$

Proof. Define

$$A_x(y) = x\operatorname{sinc}(xy) + \frac{\cos(xy)}{x}, \qquad a_x = A_x'(1) = x\cos x - 2\sin x,$$

$$b_x = A_x''(1) = (2 - x^2)\sin x - 3x\cos x, \qquad J_x(y) = \frac{y^2}{y + 1/2}\left(\frac{A_x(y) - A_x(1)}{y - 1}\right)^2,$$

with $J_x(1) = 2a_x^2/3$ and $J_x'(1) = 8a_x^2/9 + 2a_x b_x/3$ by continuity. Centering $\mathbb{E}W = 0$ and the zero-bias representation give

$$\Im(f_W(x) - f_{W^*}(x)) = \mathbb{E}[W\{A_x(Y) - A_x(1)\}].$$

By the Cauchy–Schwarz inequality and (3.3),

$$|\Im(f_W(x) - f_{W^*}(x))|^2 \le \mathbb{E}[(Y-1)^2(Y + 1/2)]\,\mathbb{E}J_x(Y) = e\,\mathbb{E}J_x(Y).$$

In Appendix A we verify the global scalar inequality

$$J_x(y) \le J_x(1) + \frac{J_x'(1)}{2}(y^2 - 1) + \frac{x^2}{4}(y-1)^2(y + 1/2) \qquad (y \ge 0,\ 0 < x \le 2). \tag{5.2}$$

Taking expectations under $Y = |W|$ and using $\mathbb{E}(Y^2 - 1) = 0$ yields $\mathbb{E}J_x(Y) \le 2a_x^2/3 + x^2 e/4$, which proves (5.1). □

5.1 Aggregation across summands by the least concave majorant

For $0 \le x \le 2$, set

$$p_1(x) = 1 + \frac{x^2}{6} - \frac{x^4}{40} + \frac{x^6}{1008}, \qquad b_0(x) = \frac{8}{3} p_1(x)^2.$$

By alternating Taylor series, $2\operatorname{sinc} x - \cos x \le p_1(x)$. Combining (4.2), (5.1), and the classical bound $|2(f_W(x) - f_{W^*}(x))/x| \le \rho = 1 + e$, we obtain

$$\left| \frac{2(f_W(x) - f_{W^*}(x))}{x} \right| \le \min\{1 + e, \max(F_1(e), F_2(e))\}, \tag{5.3}$$

where

$$F_1(e) = \sqrt{\alpha(x)^2 + b_0(x)e + e^2}, \qquad F_2(e) = \sqrt{b_0(x)e + (1 + q(x)^2)e^2},$$

since $\max\{\gamma^2, (q(x)e - \gamma)^2\} \le \max\{\alpha(x)^2, q(x)^2 e^2\}$ for $0 \le \gamma = 2g(x)/x \le \alpha(x)$.

To aggregate (5.3) across summands with different variances and third-moment excesses, we compute the explicit least concave majorant of $\max(F_1(e), F_2(e))$ on $e \ge 0$.

Lemma 5.2 (Least concave majorant of two square-root branches)**.** *Let $a, b \ge 0$, $0 \le u < v$, and $F_1(e) = \sqrt{a^2 + be + ue^2}$, $F_2(e) = \sqrt{be + ve^2}$ for $e \ge 0$. When $a = 0$, the least concave majorant of $\max(F_1, F_2)$ is F_2. When $a > 0$ and $b \le 2a\sqrt{v}$, it is the affine function $H(e) = a + \sqrt{v}\, e$. When $a > 0$ and $b > 2a\sqrt{v}$, set*

$$w = \sqrt{v - u}, \quad s = \frac{u}{w^2}, \quad r = \frac{b}{aw}, \quad d_0 = \sqrt{r^2 - 4s} > 2, \quad \eta = \frac{d_0 - 2}{r} \in (0, 1),$$

$$c = \frac{r}{2\sqrt{d_0 + s - 1}}, \quad m = \frac{d_0 + 2s}{2\sqrt{d_0 + s - 1}}, \quad e_- = \frac{a}{w}\eta, \quad e_+ = \frac{a}{w\eta}.$$

Then the least concave majorant of $\max(F_1, F_2)$ on $[0, \infty)$ is

$$H(e) = \begin{cases} F_1(e), & 0 \le e \le e_-, \\ ac + wme, & e_- \le e \le e_+, \\ F_2(e), & e \ge e_+. \end{cases}$$

Proof. For $a > 0$, rescale by $z = we/a$, so that $F_1^2/a^2 = 1 + rz + sz^2$ and $F_2^2/a^2 = rz + (s+1)z^2$, which cross at $z = 1$. When $r \le 2\sqrt{s+1}$, squaring shows that $1 + \sqrt{s+1}\, z$ majorizes both branches; since any concave majorant is at least 1 at $z = 0$ and has asymptotic slope at least $\sqrt{s+1}$ as $z \to \infty$, $1 + \sqrt{s+1}\, z$ is the least concave majorant.

When $r > 2\sqrt{s+1}$, both F_1 and F_2 are individually concave on $[0, \infty)$, and direct algebra gives the exact square completions

$$(c + mz)^2 - (1 + rz + sz^2) = c^2(z - \eta)^2,$$

$$(c + mz)^2 - \big(rz + (s+1)z^2\big) = (c^2 - 1)(z - 1/\eta)^2, \qquad c^2 - 1 = \frac{(d_0 - 2)^2}{4(d_0 + s - 1)} > 0.$$

Thus the line $c + mz$ is tangent to $\sqrt{1 + rz + sz^2}$ at $z = \eta < 1$ and tangent to $\sqrt{rz + (s+1)z^2}$ at $z = 1/\eta > 1$, and dominates both branches globally. Because H agrees with F_1 on $[0, e_-]$, with F_2 on $[e_+, \infty)$, and with the chord joining their contact points on $[e_-, e_+]$, it is concave and is dominated by every concave majorant of $\max(F_1, F_2)$. □

Applying Lemma 5.2 with $a = \alpha(x)$, $b = b_0(x)$, $u = 1$, and $v = 1 + q(x)^2$, and setting $\widehat{H}_x(e) = \min\{1 + e, H_x(e)\}$, we obtain a concave majorant $\widehat{H}_x(e)$ of the right-hand side of (5.3). For a subarray J with $\max_{i\in J} v_i \le a$, Jensen's inequality with weights $v_i^{3/2}/\tau_0$ yields

$$\sum_{i\in J} v_i |f_i(t) - f_i^*(t)| \le \frac{t}{2} P_{\mathrm{quad}}(t; B, a, \tau_0), \qquad P_{\mathrm{quad}}(t; B, a, \tau_0) = \tau_0 \widehat{H}_{t\sqrt{a}}\Big(\frac{B - \tau_0}{\tau_0}\Big), \tag{5.4}$$

where we set $P_{\mathrm{quad}} = B$ if $\tau_0 = 0$ or $t\sqrt{a} > 2$. Since $\alpha(x), b_0(x), q(x)$ are increasing on $[0, 2]$, P_{quad} is nondecreasing in t, a, and B, and because all slopes of H_x exceed 1 while $\alpha(x) < 1$, $\widehat{H}_x(e)/(1+e)$ is nondecreasing in e, which implies that P_{quad} is nonincreasing in τ_0.

As a simpler closed-form alternative for interval evaluation, note that $p_1(x) \le q(x)$ and $q(x) \ge 1$ on $[0, 2]$, so the single square-root function $R_x(e) = \sqrt{\alpha(x)^2 + \frac{8}{3}q(x)^2 e + 2q(x)^2 e^2}$ majorizes both $F_1(e)$ and $F_2(e)$ and is concave in e (since $\frac{64}{9}q^4 - 8q^2\alpha^2 > 0$). Applying Jensen's inequality to R_x gives

$$\sum_{j\in J} v_j |f_j(t) - f_j^*(t)| \le \frac{t}{2} P_{\mathrm{dir}}(t; B, a, \tau_0), \qquad t \ge 0, \tag{5.5}$$

where, with $x = t\sqrt{a}$,

$$P_{\mathrm{dir}}(t; B, a, \tau_0) = \min\left\{B, \sqrt{\alpha(x)^2\tau_0^2 + q(x)^2(B - \tau_0)(2B + 2\tau_0/3)}\right\}, \tag{5.6}$$

which likewise increases in t, a, B and decreases in τ_0 (with $P_{\mathrm{dir}} = 0$ for the empty subset).

5.2 Certified affine bounds in the third absolute moment

To complement the concave square-root bounds at moderate and large third-moment excess, we also construct certified affine bounds in $\rho = \mathbb{E}|W|^3$ for the normalized zero-bias difference

$$Z_W(x) = \frac{2(f_W(x) - f_{W^*}(x))}{x}, \qquad Z_W(0) = -\mathrm{i}\mathbb{E}W^3.$$

With $\phi_2(z) = \int_0^1 (1 - u)e^{uz}\,du$ and the entire function

$$h_x(w) = -2w^2(x + \mathrm{i}w)\phi_2(\mathrm{i}xw), \qquad h_0(w) = -\mathrm{i}w^3,$$

the zero-bias identity and $\mathbb{E}W = 0$, $\mathbb{E}W^2 = 1$ give $\mathbb{E}h_x(W) = Z_W(x)$.

Suppose that throughout a closed frequency interval $x \in [x_0, x_1]$ and for all $w \in \mathbb{R}$,

$$|h_x(w) - \mathrm{i}\beta w - \gamma(w^2 - 1)| \le c_0 + \delta + c_2|w|^2 + \Lambda|w|^3. \tag{5.7}$$

Taking expectations under W annihilates $\mathrm{i}\beta W$ and $\gamma(W^2 - 1)$, giving $|Z_W(x)| \le C + \Lambda\rho$ with $C = c_0 + c_2 + \delta$. For any common slope $\lambda \ge \Lambda > 0$, since $\rho \ge 1$, this implies $|Z_W(x)| \le (C + \Lambda - \lambda) + \lambda\rho$. Denoting by $C_\lambda(X)$ the maximum of the intercept $C + \Lambda - \lambda$ over all frequency cells in $[0, X]$, summing over $i \in J$ yields

$$\sum_{i\in J} v_i |f_i(t) - f_i^*(t)| \le \frac{t}{2}\{C_\lambda(X)\tau_0 + \lambda B\} \qquad (t\sqrt{a} \le X). \tag{5.8}$$

Two families of certificates in `certificates/scalar/` establish bounds of the form (5.8):

(i) **Modulus (disk) certificates (`disk.zip`):** Cover $x \in [0,2]$ in 128 cells of width $1/64$ at the five common slopes $\lambda \in \{5/4, 3/2, 2, 3, 4\}$. On each bounded spatial interval $|w| \le R$, outward-rounded Arb Taylor enclosures verify (5.7). For $|w| \ge R \ge 1$, positivity of the explicit cubic tail polynomial

$$(\Lambda - 1)y^3 + (c_2 - |\gamma|)y^2 - (x_1^2/2 + |\beta|)y + c_0 + \delta + |\gamma|$$

(for $\Lambda > 1$, using $|\phi_2(\mathrm{i}xy)| \le 1/2$) or

$$\Lambda y^3 + (c_2 - G)y^2 - (4/x_0^2 + |\beta + 2|)y + c_0 + \delta + G - 2/x_0$$

with $G = \max_{j\in\{0,1\}} |\gamma - 2/x_j|$ (for $0 < \Lambda \le 1$ and $x_0 > 0$, using $h_x(y) = g_x(y) - 2\mathrm{i}y - 2(1-y^2)/x$ with $|g_x(y)| \le 2/x_0 + 4y/x_0^2$) completes the verification on $\mathbb{R}$.

(ii) **Directional projection certificates (`directional.zip`):** Cover 61 frequency cells and 5 slopes across 25 directions ζ in the closed upper half-plane, which extend by complex conjugation $h_x(-w) = \overline{h_x(w)}$ to 48 directions whose convex hull contains the disk of radius $\eta = 997/1000$ (equivalently, $\max_\zeta \Re(\zeta z) \ge \eta|z|$ for all $z \in \mathbb{C}$, verified via the supporting half-planes of the 48-gon). Maximizing the intercepts over all 48 directions and dividing the entire resulting projection bound in (5.8) by η yields a second certified bound for $\sum_{i\in J} v_i|f_i(t) - f_i^*(t)|$.

We denote by $P(t; B, a, \tau_0)$ the pointwise minimum of P_{quad}, P_{dir}, and all applicable affine bounds (5.8) (with $P = 0$ for the empty subset), so that

$$\sum_{i\in J} v_i|f_i(t) - f_i^*(t)| \le \frac{t}{2} P(t; B, a, \tau_0).$$

5.3 Single-summand modulus bound from third-moment excess

Squaring the real-part bound in Lemma 3.1 and adding the square of the imaginary-part bound in Lemma 3.2 gives a second single-summand modulus bound: for a centered variable X of variance v, third absolute moment b, $\tau_j = v^{3/2}$, and $C = |\cos(t\sqrt{v})|$,

$$|f_X(t)|^2 \le C^2 + \frac{C(b - \tau_j)|t|^3}{3} + \frac{t^6(b - \tau_j)(2b + 2\tau_j/3)}{36}. \tag{5.9}$$

On any parameter box, after bounding $C = |\cos(t\sqrt{v})|$ over the cosine argument interval, the right-hand side of (5.9) is increasing in $C, b, |t|$ and decreasing in τ_j.

6 Isolating a summand of maximal variance

By the independent-sum zero-bias identity [3, Lemma 2.1(5)] at total variance one,

$$f_{S^*}(t) = \sum_{i=1}^{n} v_i f_{X_i^*}(t) \prod_{j\ne i} f_{X_j}(t). \tag{6.1}$$

Isolating the largest-variance summand X_1 (with variance $d = v_1$ and third absolute moment $b = b_1$) from the remaining summands (with parameters w, B_R, τ_R, a_R from (1.5)), we bound the remainder product by

$$R(t) = \min\{e^{-\psi(t;w,B_R+\tau_R)},\ \mathcal{F}_0(t; w, B_R, a_R, \tau_R)\},$$

the single-summand modulus $|f_1(t)|$ by $A(t)$, the minimum of (2.4) and the square root of (5.9), the deleted remainder product by $D_R(t) = \mathcal{D}_0(t; w, B_R, a_R, \tau_R)$ (if the remainder is empty, we set

$R(t) = D_R(t) = 1$ and $P(t; B_R, a_R, \tau_R) = 0$), and the deleted whole-sum product by $D_S(t) = \mathcal{D}_0(t; 1, L, d, \tau)$. Splitting (6.1) into $i = 1$ and $i \geq 2$ gives

$$|f_S(t) - f_{S^*}(t)| \leq \frac{t}{2}\Big\{P(t; b, d, d^{3/2})R(t) + P(t; B_R, a_R, \tau_R)A(t)D_R(t)\Big\}. \tag{6.2}$$

Combining (6.2) with the unsplit bound $(t/2)P(t; L, d, \tau)D_S(t)$ and the classical bound $tL/2$ yields $|f_S(t) - f_{S^*}(t)| \leq (tL/2)\mathcal{B}(t)$, where

$$\mathcal{B}(t) = \min\left\{1,\ \frac{P(t; L, d, \tau)D_S(t)}{L},\ \frac{P(t; b, d, d^{3/2})R(t) + P(t; B_R, a_R, \tau_R)A(t)D_R(t)}{L}\right\}. \tag{6.3}$$

Similarly, for the full characteristic function $|f_S(t)|$, we use

$$F_S(t) = \min\big\{e^{-\psi(t;1,L+\tau)},\ A(t)R(t),\ \mathcal{F}_0(t; 1, L, d, \tau)\big\}. \tag{6.4}$$

Notice that the third expression in (6.3) can also be written as the convex combination $\lambda U_1 + (1 - \lambda)U_R$, where $\lambda = b/L \in [0, 1]$, $U_1 = P(t; b, d, d^{3/2})R(t)/b \in [0, 1]$, and $U_R = P(t; B_R, a_R, \tau_R)A(t)D_R(t)/B_R \in [0, 1]$ (setting $U_R = 0$ when $B_R = 0$). On a parameter box where $\lambda \in [\lambda_-, \lambda_+]$ and $U_1 \leq \overline{U}_1$, $U_R \leq \overline{U}_R$, we have

$$\lambda U_1 + (1 - \lambda)U_R \leq \max_{\lambda \in \{\lambda_-, \lambda_+\}} \{\lambda \overline{U}_1 + (1 - \lambda)\overline{U}_R\}, \tag{6.5}$$

and the evaluator takes the minimum of (6.5) and direct interval division of the numerator in (6.3) by L_-.

Finally, integrating $(f_S(t) - e^{-t^2/2})' + t(f_S(t) - e^{-t^2/2}) = t(f_S(t) - f_{S^*}(t))$ as in Tyurin [13, Lemma 3] yields the modulus bound

$$\frac{|f_S(t) - e^{-t^2/2}|}{L} \leq R_\Delta(t) := e^{-t^2/2} \int_0^t \frac{u^2}{2}\mathcal{B}(u)e^{u^2/2}\, du. \tag{6.6}$$

7 Signed comparison with a product of cosines

To preserve the sign of the characteristic-function displacement when the summands are close to symmetric two-point distributions, we compare $f_S(t)$ with the reference cosine product

$$C_v(t) = \prod_{i=1}^{n} \cos(t\sqrt{v_i}),$$

which is the characteristic function of $\sum_{i=1}^n \sqrt{v_i}\varepsilon_i$ for independent Rademacher variables ε_i [6]. Put $g(t) = e^{-t^2/2}$, $k_v(t) = \sum_{i=1}^n \sqrt{v_i}\tan(t\sqrt{v_i})$, and $r_i(t) = f_i'(t) + \sqrt{v_i}\tan(t\sqrt{v_i})f_i(t)$. For $t\sqrt{d} < \pi/2$, every cosine factor is positive, and differentiating $f_S(t) - C_v(t)$ by the product rule yields the exact linear differential identity

$$(f_S - C_v)'(t) + k_v(t)(f_S(t) - C_v(t)) = \sum_{i=1}^{n} r_i(t) \prod_{j \neq i} f_j(t), \qquad f_S(0) - C_v(0) = 0. \tag{7.1}$$

7.1 Bounding the forcing term

For a standardized summand W of mean zero and variance one, and $0 < x < \pi/2$, define

$$T_x(w) = \frac{2}{x^2}\{(\mathrm{i}w + \tan x)e^{\mathrm{i}xw} - \mathrm{i}w\sec x - x(1-w^2)\}, \qquad T_0(w) = \mathrm{i}(w - w^3). \tag{7.2}$$

Since $\mathbb{E}W = 0$ and $\mathbb{E}W^2 = 1$, we have

$$\mathbb{E}T_x(W) = \frac{2\{f_W'(x) + \tan(x) f_W(x)\}}{x^2}, \qquad r_i(t) = \frac{1}{2}t^2 v_i^{3/2}\mathbb{E}T_{t\sqrt{v_i}}(W_i).$$

An elementary Wronskian bound holds on the entire interval $0 \le x < \pi/2$: writing $q_0(u) = f_W''(u) + f_W(u) = \mathbb{E}[(1-W^2)(e^{\mathrm{i}uW} - 1)]$, the Cauchy–Schwarz inequality from the proof of Lemma 3.2 gives $|q_0(u)| \le u\sqrt{(\rho-1)(\rho+5/3)}$. Since $f_W'(x)\cos x + f_W(x)\sin x = \int_0^x q_0(u)\cos u\, du$ and $\cos u > 0$, we obtain

$$\begin{aligned} |\mathbb{E}T_x(W)| &\le 2p(x)\sqrt{(\rho-1)(\rho+5/3)} \quad (0 \le x < \pi/2), \\ p(x) &= \frac{x\tan x + 1 - \sec x}{x^2} = \int_0^1 \frac{r\cos(rx)}{\cos x}\, dr, \quad p(0) = \frac{1}{2}. \end{aligned} \tag{7.3}$$

On the subinterval $0 \le x \le 1$, Appendix B establishes the sharper quadratic inequality

$$|\mathbb{E}T_x(W)|^2 \le (\rho-1)\left\{q_{\mathrm{ref}}(x) + \frac{9}{8}(\rho-1)\right\}, \qquad q_{\mathrm{ref}}(x) = \frac{8}{3}\frac{\operatorname{sinc}^4 x}{\cos^2 x} = \frac{8}{3}\left(\frac{\sec x - \cos x}{x^2}\right)^2. \tag{7.4}$$

Let $D_{\mathrm{del}}(t)$ be an upper bound on $\max_i \prod_{j\ne i}|f_j(t)|$. Applying Cauchy–Schwarz with $w_i = v_i^{3/2}$ and $e_i = b_i - w_i$ gives, for $t\sqrt{d} \le 1$,

$$\sum_{i=1}^n w_i |\mathbb{E}T_{t\sqrt{v_i}}(W_i)| \le \sqrt{(L-\tau)\left\{\sum_{i=1}^n w_i q_{\mathrm{ref}}(t w_i^{1/3}) + \frac{9}{8}(L-\tau)\right\}}.$$

Because $\sec x$ has nonnegative even Taylor coefficients with $[\sec x]_{x^{2n}} \ge 1/(2n)!$, the function $f_t(w) = w q_{\mathrm{ref}}(t w^{1/3}) = \sum_{n\ge 0} c_n t^{2n} w^{1+2n/3}$ has $c_n \ge 0$ and is increasing and convex on $w \ge 0$. Subject to $0 \le w_i \le a = d^{3/2}$ and $\sum_i w_i = \tau$, a convex sum $\sum_i f_t(w_i)$ is maximized by packing as many weights as possible at the upper cap a: setting $k = \lfloor \tau/a \rfloor$ and $r = \tau - ka$, we have $\sum_{i=1}^n w_i q_{\mathrm{ref}}(t w_i^{1/3}) \le F_t(a,\tau) := k f_t(a) + f_t(r)$. Multiplying by $t^2 D_{\mathrm{del}}(t)/2$ yields the packed forcing bound

$$\left|\sum_{i=1}^n r_i(t)\prod_{j\ne i} f_j(t)\right| \le \frac{t^2 D_{\mathrm{del}}(t)}{2}\sqrt{(L-\tau)\left\{F_t(a,\tau) + \frac{9}{8}(L-\tau)\right\}}. \tag{7.5}$$

7.2 Sharpening the forcing bound via the largest summand

When the third absolute moment b of the largest-variance summand is tracked separately, write

$$w = d^{3/2}, \quad e = b - w, \quad R = \tau - w, \quad e_R = L - b - R, \quad E = L - \tau = e + e_R, \quad \delta = Lw - b\tau.$$

For any scalar bound of the form $|\mathbb{E}T_x(W_i)|^2 \le (\rho_i - 1)\{Q + \lambda(\rho_i - 1)\}$ with $Q > 0$ and $\lambda \ge 0$, applying Cauchy–Schwarz only over the remaining summands $i \ge 2$ gives the split moment factor

$$S_Q = \sqrt{e(Qw + \lambda e)} + \sqrt{e_R(QR + \lambda e_R)}. \tag{7.6}$$

Setting $A = Qw+\lambda e$, $B = QR+\lambda e_R$, and $G^2 = E(Q\tau+\lambda E)$, the identity $G^2-S_Q^2 = (\sqrt{eB}-\sqrt{e_RA})^2$ together with $eB - e_RA = -Q\delta$ and $(\sqrt{eB}+\sqrt{e_RA})^2 \le G^2$ implies that when $\delta > 0$,

$$S_Q \le G\sqrt{1-\left\{\frac{\delta}{E(\tau+(\lambda/Q)E)}\right\}^2}. \tag{7.7}$$

Similarly, for the convex weight bound with $q = q_{\mathrm{ref}}$, $\lambda = 9/8$, $q_j = q(tw^{1/3}) \ge 8/3$, and $F_R = F_t(w,R)$, the split factor

$$S_P = \sqrt{e(q_jw+\lambda e)} + \sqrt{e_R(F_R+\lambda e_R)} \tag{7.8}$$

satisfies $F_t(w,\tau) = wq_j + F_R$ and $F_R \le q_jR$, so that $eB - e_RA \le -q_j\delta$ when $\delta \ge 0$, yielding

$$S_P \le \sqrt{E\left\{F_t(w,\tau)+\frac{9}{8}E\right\}}\sqrt{1-\left\{\frac{\delta}{E(\tau+27E/64)}\right\}^2} \qquad (\delta \ge 0). \tag{7.9}$$

The same contraction factor applies to (7.7) whenever $\lambda/Q \le 27/64$.

To bound (7.6) and (7.8) on a parameter box, choose $w_- \le d_-^{3/2}$ and $w_+ \ge d_+^{3/2}$, and set $M = L_+$ and $a = \tau_-$. Suppose

$$a - w_+ \ge \frac{15}{23}(M - b_-). \tag{7.10}$$

With w, b fixed, both S_Q and S_P increase with $L \le M$. Putting $H = M - b$, the derivative of the squared remainder term with respect to R is

$$\frac{d}{dR}\{(H-R)[QR+\lambda(H-R)]\} = Q(e_R-R) - 2\lambda e_R \le 0 \qquad (R \ge H/2),$$

while for S_P, since $F_t(w,R) \ge 8R/3$ and $0 \le \partial_R F_t(w,R) < 29/4$ between multiples of w,

$$\frac{d}{dR}\left\{(H-R)\left[F_t(w,R)+\frac{9}{8}(H-R)\right]\right\} \le 5e_R - \frac{8}{3}R \le 0 \qquad (R \ge 15H/23).$$

Thus (7.10) implies $S_Q(L,\tau,w,b) \le S_Q(M,a,w,b)$ and $S_P(L,\tau,w,b) \le S_P(M,a,w,b)$. With $E_* = M-a$, $\delta_* = Mw_- - b_+a$, and $D_* = E_*(a+27E_*/64)$, any rational $\eta \in (0,1]$ satisfying $\eta^2+(\delta_*/D_*)^2 \ge 1$ for $0 < \delta_* \le D_*$ yields

$$S_Q(L,\tau,w,b) \le \eta\sqrt{E_*(Qa+\lambda E_*)}, \qquad S_P(L,\tau,w,b) \le \eta\sqrt{E_*\{F_t(w,a)+9E_*/8\}}.$$

When the remaining weight $R = \tau - w$ is smaller than w throughout the box, a relaxed endpoint condition suffices for S_P: with $R_- = a - w_+$, $R_+ = \tau_+ - w_-$, and $0 < W_0 \le d_+^{3/2}$, if

$$0 < R_- \le R_+ < w_-, \qquad R_+ \le (19/25)^3W_0, \qquad R_- \ge \frac{23}{55}(M-b_-), \tag{7.11}$$

then $F_t(w,R) = f_t(R)$ with $tR^{1/3} \le 19/25$, where $f_t'(R) < 25/6$ gives $\frac{d}{dR}\{e_R[f_t(R)+\frac{9}{8}e_R]\} \le \frac{23}{12}e_R - \frac{8}{3}R \le 0$. Both derivative bounds $\partial_R F_t < 29/4$ (for $x \le 10001/10000$) and $f_t'(R) < 25/6$ (for $x \le 19/25$) follow by bounding the increasing function

$$g(x) = q(x) + \frac{1}{3}xq'(x) = \frac{8\sin^3 x\{2x(1+\cos^2 x) - \sin x\cos x\}}{9x^4\cos^3 x}$$

via alternating Taylor polynomials of $\sin x$ and $\cos x$.

7.3 Joint bound on the convex weight maximum and Cauchy–Schwarz defect

Because $F_t(w,a)$ increases with w while the defect $\delta = Mw - ba$ vanishes at $w = ba/M$, their maxima cannot occur simultaneously when ba/M is strictly smaller than the weight cap $W \ge d_+^{3/2}$. Specifically, suppose (7.10) or (7.11) holds, set $E = M - a > 0$ and $\lambda = 9/8$, and choose c, W_0 with $b_+a/M < c < W_0 \le W$ such that $\lfloor a/u \rfloor = k \ge 1$ on (c, W_0). Choose $d_0 > \gamma \ge 0$ with $u^{2/3} \ge d_0 > \gamma \ge (a-ku)^{2/3}$ on $[c, W_0]$. Using the Taylor expansion $q(x) = \sum_{n\ge 0} q_n x^{2n}$ with $q_0 = 8/3$, $q_1 = 8/9$, $q_2 = 8/15$, $q_3 = 736/2835$, and $q_n \ge 0$, we have for any $N \ge 1$,

$$\frac{d}{du}F_t(u,a) = k\{f_t'(u) - f_t'(a-ku)\} \ge k\sum_{n=1}^{N} q_n(1+2n/3)(d_0^n - \gamma^n)t^{2n}.$$

On a frequency cell $[l, r]$ with $r\sqrt{d_+} \le 1$ and $rW^{1/3} \le 10001/10000$, choose $\bar{q} \ge q(rW^{1/3})$ and set

$$\begin{aligned} A &= \min\{(29/4)a + \lambda E,\ a\bar{q} + \lambda E\}, \\ P_N(y) &= \frac{k(W_0 - c)}{A}\sum_{n=1}^{N} q_n(1+2n/3)(d_0^n - \gamma^n)y^n, \\ H &= \left\{\frac{Mc - b_+a}{E(a + 27E/64)}\right\}^2 \le 1. \end{aligned}$$

Integrating $\frac{d}{du}F_t(u,a)$ over $[c, W_0]$ bounds S_P^2 for $w \le c$ by $\{1 - P_N(t^2)\}E\{F_t(W,a) + \lambda E\}$, while for $w \ge c$ the defect bound (7.9) gives the factor $1 - H$. Using $\sqrt{1-x} \le 1 - x/2$, we obtain across the entire box and frequency cell

$$S_P(L, \tau, w, b) \le \left(1 - \frac{1}{2}\min\{P_N(l^2), H\}\right)\sqrt{E\{F_t(W,a) + \lambda E\}}. \tag{7.12}$$

7.4 Enclosing the cosine product and integrating the differential recurrence

Lemma 7.1. *Let $M = \sqrt{d} < 1$ and $m = (\tau - M^3)/(1 - M^2)$. For every increasing convex function h on $[0, M]$,*

$$\sum_{i=1}^{n} v_i h(\sqrt{v_i}) \ge M^2 h(M) + (1 - M^2)h(m).$$

In particular, for $0 \le l \le r$ with $rM < \pi/2$,

$$\begin{aligned} C_v(t) &\le C_*(t) := \cos(tM)\cos(tm)^{(1-M^2)/m^2}, \\ k_v(t) &\ge K_*(t) := M\tan(tM) + (1 - M^2)\frac{\tan(tm)}{m}, \\ \frac{C_v(r)}{C_v(l)} &\le \frac{C_*(r)}{C_*(l)}. \end{aligned}$$

Proof. Isolating one summand of variance $M^2 = d$ and applying Jensen's inequality with weights $v_i/(1 - M^2)$ to the remaining summands (whose weighted mean of $\sqrt{v_i}$ is $m \in [0, M]$) gives the first inequality. Applying it to the increasing convex functions $-\log\cos(tx)/x^2$, $\tan(tx)/x$, and $[\log\cos(lx) - \log\cos(rx)]/x^2$ yields the three cosine bounds. □

On a parameter box, setting $D_0 = \max\{d_{lo}, \tau_{lo}^2\}$, $M_0 = \sqrt{D_0}$, $\tau_0 = \max\{\tau_{lo}, M_0^3\}$, and $m_0 = (\tau_0 - M_0^3)/(1 - D_0)$ (when $D_0 < 1$), the lower corner (M_0, τ_0) minimizes $M^2h(M) + (1 - M^2)h(m)$

over the feasible box because both partial derivatives ∂_M and ∂_τ are nonnegative. Writing $J(z) = -\log\cos z - z^2/2$ and $A_*(t) = D_0 J(tM_0)/M_0^2 + (1-D_0)J(tm_0)/m_0^2$, the chord upper bound for $J(tx)/x^2$ on $[0,\sqrt{d}]$ and Lemma 7.1 enclose the real displacement $C_v(t) - g(t)$ on $[l,r]$ within

$$g(l)\,\mathrm{expm1}\left[-\frac{\tau_{hi}}{d_{hi}^{3/2}}J(r\sqrt{d_{hi}})\right] \le C_v(t) - g(t) \le g(r)\,\mathrm{expm1}[-A_*(l)] \le 0. \tag{7.13}$$

Finally, if A_j bounds the forcing right-hand side of (7.1) on $[l,r]$, P_j bounds $|f_S(l) - C_v(l)|$, k_j is a lower bound for k_v on $[l,r]$, and Q_j bounds $C_v(r)/C_v(l)$, then integrating (7.1) with $h = r-l$ and $\Psi_j = (1-e^{-k_j h})/k_j$ gives the recurrence

$$E_j = \max\{P_j, e^{-k_j h}P_j + A_j\Psi_j\}, \qquad P_{j+1} \le Q_j P_j + A_j\Psi_j, \tag{7.14}$$

where E_j bounds $|f_S(t) - C_v(t)|$ throughout $[l,r]$ and P_{j+1} bounds it at the right endpoint r.

8 Prawitz smoothing and threshold partitions

For $0 < u < 1$, Prawitz's smoothing kernel is

$$K(u) = \frac{1-u}{2} + \frac{\mathrm{i}}{2}\left((1-u)\cot(\pi u) + \frac{1}{\pi}\right).$$

In its unsigned form [9, 12, 13], for any cutoff $T > 0$ and split point $0 < s < 1$,

$$\frac{\Delta}{L} \le 2\int_0^s |K(u)|R_\Delta(Tu)\,du + \frac{2\int_s^1 |K(u)|F_S(Tu)\,du + G(T,s)}{L}, \tag{8.1}$$

where $E_1(z) = \int_z^\infty v^{-1}e^{-v}\,dv$ and

$$G(T,s) = 2\int_0^s \left|K(u) - \frac{\mathrm{i}}{2\pi u}\right| e^{-T^2u^2/2}\,du + \frac{E_1(T^2s^2/2)}{2\pi}. \tag{8.2}$$

8.1 Cell integration and the Gaussian correction

On a frequency subinterval $[a,b] \subseteq [0,Ts]$, let $\mathcal{B}_j$ bound $\mathcal{B}(u)$ on $[a,b]$, let $I_a = \int_0^a \frac{1}{2}u^2\mathcal{B}(u)e^{u^2/2}\,du$, and let A_j bound $2\pi u|K(u)|$ on $[a/T, b/T]$. Splitting the integral defining $R_\Delta(t)$ at a, bounding $e^{-(t^2-u^2)/2} \le 1$ on $a \le u \le t$, and interchanging the order of integration bounds the contribution of $[a,b]$ to the first term of (8.1) by

$$\frac{A_j}{\pi}\left\{I_a\frac{E_1(a^2/2) - E_1(b^2/2)}{2} + \mathcal{B}_j\left(\frac{b^3-a^3}{18} - \frac{a^3}{6}\log\frac{b}{a}\right)\right\}, \tag{8.3}$$

where for $a > 0$ we take the minimum with the direct integral of $|f_S(t) - e^{-t^2/2}| \le F_S(t) + e^{-t^2/2}$.

For the Gaussian term $G(T,s)$, subtracting the pole $\mathrm{i}/(2\pi u)$ gives

$$2\left|K(u) - \frac{\mathrm{i}}{2\pi u}\right| = (1-u)\sqrt{1+h(u)^2}, \qquad h(u) = \frac{1}{\pi u} - \cot(\pi u) = \frac{2u}{\pi}\sum_{n=1}^{\infty}\frac{1}{n^2-u^2} \tag{8.4}$$

[7, eq. 4.22.3]. Truncating the partial-fraction series at $M = 8$ with remainder cap $c_M(s) = (\pi^2/6 - \sum_{n=1}^M n^{-2})/(1-s^2/(M+1)^2)$ gives the smooth upper integrand $h_M(u) = (2u/\pi)(\sum_{n=1}^M (n^2-u^2)^{-1} + c_M(s))$, so that

$$G(T,s) \le \int_0^s (1-u)\sqrt{1+h_M(u)^2}e^{-T^2u^2/2}\,du + \frac{E_1(T^2s^2/2)}{2\pi}, \tag{8.5}$$

which is enclosed with Arb alongside the elementary bound of Shevtsova [12, Lemma 4.1]:

$$G(T,s) \le \int_0^s \left(1 - u + \frac{\pi^2 u^2}{18}\right) e^{-T^2u^2/2}\,du + \frac{E_1(T^2s^2/2)}{2\pi}. \tag{8.6}$$

8.2 Signed smoothing over a threshold partition

With $k_T(t) = K(t/T)/T$ and $g(t) = e^{-t^2/2}$, the signed smoothing inequality of Prawitz [9] (see also Pinelis [8, eqs. (2.2)–(2.6)]) states that for every $x \in \mathbb{R}$,

$$\mathbb{P}\{S \le x\} \le \frac{1}{2} + 2\Re \int_0^T k_T(t) e^{-\mathrm{i}xt} f_S(t)\,dt. \tag{8.7}$$

Define the signed Gaussian defect at intermediate cutoff $a \in (0, Ts]$ by

$$\Gamma_{T,a}(x) = \frac{1}{2} + 2\Re \int_0^a k_T(t) e^{-\mathrm{i}xt} g(t)\,dt - \Phi(x).$$

On the frequency mesh $t_j = Tsj/N$ $(0 \le j \le N)$, let C_j bound $(2/L)\int_{t_j}^{t_{j+1}} |k_T(t)|\,|f_S(t) - g(t)|\,dt$, let A_j bound $2\int_{t_j}^{t_{j+1}} |k_T(t) f_S(t)|\,dt$, and let H bound $2\int_{Ts}^{T} |k_T(t) f_S(t)|\,dt$. Splitting (8.7) at t_k $(1 \le k \le N)$ yields

$$\frac{\mathbb{P}\{S \le x\} - \Phi(x)}{L} \le \sum_{j<k} C_j + \frac{\sum_{j\ge k} A_j + H + \Gamma_{T,t_k}(x)}{L}. \tag{8.8}$$

When x is restricted to a closed threshold interval $I \subset \mathbb{R}$, the cosine-product comparison of Section 7 sharpens the cell bound on $[l, r] = [t_j, t_{j+1}]$ $(j \ge 1)$. Let $[\alpha, \beta]$ enclose $\Re(k_T(t)e^{-\mathrm{i}xt})$ for $(t, x) \in [l, r] \times I$. If $|f_S - C_v| \le E_j$ and $C_v - g \in [\ell_j, h_j]$ on $[l, r]$, then

$$2\Re \int_l^r k_T(t) e^{-\mathrm{i}xt}(f_S(t) - g(t))\,dt \le J_{j,I} := W_j E_j + 2(r-l)\max\{\ell_j\alpha, \ell_j\beta, h_j\alpha, h_j\beta\}, \tag{8.9}$$

where $W_j \ge 2\int_l^r |k_T(t)|\,dt$. Because $C_v - g \le 0$, cells on which $\Re(k_T(t)e^{-\mathrm{i}xt}) > 0$ contribute a negative drift term to $J_{j,I}$. For $L \in [L_{lo}, L_{hi}]$, defining $\mathcal{N}(z) = z/L_{lo}$ for $z \ge 0$ and $\mathcal{N}(z) = z/L_{hi}$ for $z < 0$ bounds $J_{j,I}/L$ by $\mathcal{N}(J_{j,I})$.

Furthermore, on any frequency–threshold rectangle $[l, r]\times I$ where $|D| \le R$ (with $D = (f_S - g)/L$) and $|f_S| \le r$, any choice of a rational complex multiplier $h \in \mathbb{Q}[\mathrm{i}]$ gives the two-disk support inequality

$$\Re(kD) = \Re((k-h)D) + \frac{\Re(hf_S) - g\Re h}{L} \le R|k-h| + \mathcal{N}(r|h| - g\Re h), \qquad k = k_T(t)e^{-\mathrm{i}xt}, \tag{8.10}$$

which interpolates between the discrepancy bound ($h = 0$) and the modulus bound ($h = k$). Letting $B_{c,j,I}$ denote the minimum of C_j, $\mathcal{N}(J_{j,I})$, and any certified two-disk bound (8.10) for smoothing profile c, and letting $\gamma_{c,k,I}$ bound $\sup_{x\in I} \Gamma_{T_c,t_k}(x)$, we obtain on I

$$U_{c,k,I} = \sum_{j<k} B_{c,j,I} + \mathcal{N}\left(\sum_{j\ge k} A_{c,j} + H_c + \gamma_{c,k,I}\right).$$

Covering $[-X, X]$ (with $X = 2$ or $X = 69/32$) by a finite partition of closed intervals I and bounding $|x| \ge X$ by Cantelli's inequality gives

$$\frac{\Delta}{L} \le \max\left\{\max_I \min_{c,k} U_{c,k,I},\ \frac{\Delta_{\mathrm{ext}}(X)}{L_{lo}}\right\}, \tag{8.11}$$

where $\Delta_{\mathrm{ext}}(X) \le 1/(1+X^2)$, and for $X = 2$ the refined Cantelli–normal tail bound gives

$$\Delta_{\mathrm{ext}}(2) = \sup_{|x|\ge 2} |\mathbb{P}(S \le x) - \Phi(x)| \le \frac{1}{5} - (1 - \Phi(2)) < 0.17724986805182083, \tag{8.12}$$

since $h(x) = (1+x^2)^{-1} - (1-\Phi(x))$ dominates $1-\Phi(x)$ and decreases on $[2,\infty)$. Reflection $S \mapsto -S$ preserves the moment box, the even reference function C_v, and the modulus bounds, so (8.11) bounds both one-sided Kolmogorov discrepancies.

8.3 Enclosing the signed Gaussian defect $\Gamma_{T,a}(x)$

With $a = Ts$ and $h(u) = 1/(\pi u) - \cot(\pi u)$, $\Gamma_{T,a}(x)$ has the exact representation

$$\Gamma_{T,a}(x) = \int_0^s (1-u)e^{-T^2u^2/2}\{\cos(xTu) - h(u)\sin(xTu)\}\,du - \frac{1}{\pi}\int_s^\infty e^{-T^2u^2/2}\frac{\sin(xTu)}{u}\,du.$$

Approximating $h(u)$ by its Taylor polynomial $p_m(u) = \sum_{j=0}^{m-1}(2\zeta(2j+2)/\pi)u^{2j+1}$ incurs an integral error at most $\epsilon_m = \pi s^{2m+2}/[3(1-s^2)(2m+2)]$ and reduces the main term to the entire integrand

$$\Gamma^{(m)}(x) = -\frac{1}{2}\operatorname{erf}(x/\sqrt{2}) + \int_0^a g(t)\left[\frac{1-t/T}{T}\{\cos(xt) - p_m(t/T)\sin(xt)\} + \frac{x}{\pi}\operatorname{sinc}(xt)\right]dt,$$

which is enclosed at grid nodes using Arb's Petras integration algorithm [5]. Between adjacent threshold nodes $u < v$ with upper enclosures G_u, G_v, the global second-derivative bound

$$|\Gamma''_{T,a}(x)| \le M_G := \left(1 + \frac{\pi s}{3(1-s^2)}\right)\left\{\frac{\sqrt{\pi/2}}{T}\operatorname{erf}(a/\sqrt{2}) - se^{-a^2/2}\right\} + \frac{e^{-a^2/2}}{\pi}$$

yields the concave quadratic upper envelope

$$\Gamma_{T,a}(x) \le \frac{(v-x)G_u + (x-u)G_v}{v-u} + \frac{M_G}{2}(x-u)(v-x) \qquad (u \le x \le v). \tag{8.13}$$

9 The certified interval calculation

The interval cover spans $L \in [L_-, L_+]$, with exact binary64 rational endpoints

$$L_- = \frac{6456360425798343}{4611686018427387904} \approx 0.0014, \qquad L_+ = \frac{1362338887279575}{1125899906842624} \approx 1.21. \tag{9.1}$$

9.1 Analytic bounds for small and large L

For $0 < L \le L_- \le 0.01$, Corollary 4.18 of Shevtsova [12] gives

$$\frac{\Delta}{L} \le C_{\mathrm{E}} + 0.3413L^{1/3} \le C_{\mathrm{E}} + 0.3413L_-^{1/3} < 0.447913037296. \tag{9.2}$$

For $L \ge L_+$, Cantelli's inequality and reflection give, for every mean-zero, variance-one random variable S,

$$\Delta \le \sup_{x>0}\left(\Phi(x) - \frac{x^2}{1+x^2}\right) < 0.540936541549, \tag{9.3}$$

where the unique stationary point of $\Phi(x) - x^2/(1+x^2)$ on $(0,\infty)$ is enclosed with Arb. Dividing by L_+ gives

$$\frac{\Delta}{L} < 0.447055000001 \qquad (L \ge L_+). \tag{9.4}$$

9.2 Partition of the intermediate moment domain

On each closed band $L \in [\ell, H]$ of the 1,670-band cover of $[L_-, L_+]$, the root box $0 \le d \le \min\{1, H^{2/3}\}$, $0 \le b \le H$, $0 \le \tau \le \min\{1, H\}$ is recursively bisected into closed subboxes. Each leaf box is either certified infeasible by (1.4) or certified to satisfy $\Delta/L \le U_*$ by outward-rounded interval evaluation (including, on 10,238 leaves, a complete nested partition of the feasible b or d interval).

Quantity	Recorded value
Closed L bands covering $[L_-, L_+]$	1,670
Outer partition tree nodes	483,030
Accepted outer leaves	235,659
Infeasible outer leaves	6,691
Individually evaluated accepted leaves	143,618
Containing-band-bounded accepted leaves	92,041
Largest certified upper bound U_* on $[L_-, L_+]$	$\approx 0.44987999580838745$
Analytic upper bound for $0 < L \le L_-$	< 0.447913037296
Analytic upper bound for $L \ge L_+$	< 0.447055000001

The maximum across all accepted leaves is the exact dyadic rational

$$U_* = \frac{1013039690742047}{2251799813685248} < 0.44988. \tag{9.5}$$

Since both analytic complements (9.2) and (9.4) are strictly smaller than U_*, Theorem 1.1 follows.

9.3 Arithmetic conventions and verification mapping

All vector evaluations use IEEE-754 binary64 arithmetic with outward rounding via `nextafter`, an upward summation allowance of $1 + 4N \cdot 2^{-53}$ for N-term nonnegative sums, and the downward-evaluated rational exponential bound $e^{-p} \le P_{16}(p/64)^{-64}$ for $0 \le p < 25$, where $P_{16}(x) = \sum_{j=0}^{16} x^j/j!$. Scalar constants, logarithmic tangent checks, affine certificates, and Gaussian node integrals use Arb ball arithmetic [4, 5].

The correspondence between the scalar lemmas of this paper and the verification functions in `verify/check_scalar_certificates.py` is summarized below:

Mathematical estimate	Certificate files in `certificates/scalar/`	Verification function
Logarithmic tangent bounds, Lemmas 2.2–2.3	`direct-log.json`	`check_direct_log`
Quadratic imaginary-error bound, Lemma 5.1 & Appendix A	`odd-quadratic.json`	`check_odd`
Affine zero-bias bounds, (5.7)–(5.8)	`affine-index.json`, `disk.zip`, `directional.zip`	`check_affine`
Cosine-comparison forcing bound, (7.4) & Appendix B	`energy.json`	`check_energy`

The moment-partition cover and Gaussian defect tables are verified by `verify/check_saved.py` and `verify/check_replays.py`, as documented in `verify/README.md` and `certificates/scalar/README.md`.

A Proof of the scalar majorant for the imaginary error

To prove (5.2) for $0 < x \le 2$ and $y \ge 0$, set $z = xy$, $q = 1 + x^2/3$, and

$$R(x,z) = \frac{\cos z + x^2 \operatorname{sinc} z - \cos x - x^2 \operatorname{sinc} x}{z^2 - x^2},$$

extended by continuity across $z = x$. Multiplying the difference between the right- and left-hand sides of (5.2) by $x(z + x/2)$ gives

$$\mathcal{L}(x,z) = x(z+x/2)\left\{J_x(1) + \frac{J_x'(1)}{2x^2}(z^2 - x^2)\right\} + \frac{(z-x)^2(z+x/2)^2}{4} - z^2(z+x)^2 R(x,z)^2,$$

which has a double zero at $z = x$. Put

$$w_n(u) = \frac{1}{(2n)!} + \frac{u}{(2n+1)!}, \qquad S_n(x,z) = \sum_{j=0}^{n-1} x^{2j} z^{2(n-1-j)}.$$

For $r = m + n$, the homogeneous polynomial of degree $2r - 2$ defined by exact division,

$$Q_{mn}(x,z) = \frac{z^2(z+x)^2 S_m(x,z) S_n(x,z) - \frac{4mn}{9} x^{2r-3}(z+x/2)\{(5-3r)x^2 + (3r+1)z^2\}}{(z-x)^2},$$

has positive rational coefficients and satisfies $Q_{mn}(1,1) \le 8mnr^2$. Expanding $R(x,z)$ in powers of x and z yields the power-series identity

$$\frac{\mathcal{L}(x,z)}{(z-x)^2} = \frac{q^2 x(x+4z)}{144} - \frac{(q^2-1)(z+x/2)^2}{4} + \sum_{r \ge 3} (-1)^{r+1} \sum_{m=1}^{r-1} w_m(x^2) w_{r-m}(x^2) Q_{m,r-m}(x,z). \tag{A.1}$$

Truncating (A.1) at $r = 24$ and decomposing the resulting polynomial into parity components as $x^2 F_1(x^2, z^2) + z^4 F_0(z^2) + xz G(x^2, z^2)$ on $(x^2, z^2) \in [0,4] \times [0,49]$, exact tensor Bernstein conversion gives minimum Bernstein coefficients of $1/144$ for F_1, $> 1/200$ for F_0, and $1/36$ for G. Using $w_n(x^2) \le 7/[3(2n)!]$ and $\sum_{m=1}^{r-1} m(r-m)/[(2m)!(2r-2m)!] = 2^{2r-5}/(2r-2)!$, the degree-$r$ term at $(x,z) = (2,7)$ is bounded by $B_r = (49/1764) r^2 196^r/(2r-2)!$, so each component tail for $r \ge 25$ is at most $5B_{25}/18 < 1/1250$. Subtracting this tail leaves each component strictly above $1/250$, proving $\mathcal{L}(x,z) \ge 0$ for $0 \le z \le 7$.

For $z \ge 7$, the bound $|A_x(y) - A_x(1)| \le 2(x + 1/x)$ gives

$$\frac{J_x(y)}{(x^2/4)h(y)} \le \frac{16(1+x^2)^2 z^2}{(z-x)^4 (z+x/2)^2} \le \frac{400}{(z-2)^4} \le \frac{16}{25} < 1,$$

while $J_x(1) \ge 0$ and $J_x'(1) = 2x^2 k_0(4k_0 - 3\ell_0)/9 > 0$ on $(0,2]$ (with $k_0 = 2\operatorname{sinc} x - \cos x > 0$, $\ell_0 = (2 - x^2)\operatorname{sinc} x - 3\cos x$, and $4k_0 - 3\ell_0 \ge 257/105$). Thus (5.2) holds for all $y \ge 0$.

B Proof of the scalar bound for the cosine comparison

To prove (7.4) on $0 \le x \le 1$, set for $x > 0$

$$a_x = \frac{2x - x\sec x - \sin x}{x^2}, \qquad U_x(w) = T_x(w) - a_x(w^2 - 1), \qquad V_x(y) = \frac{x^2 \cos x}{2} U_x(y).$$

Writing $s = \sin x$, $c = \cos x$, $A = (x + sc)/2$, $B = s^2$, and $D = x^2 + 2xsc$, we have $V_x(y) = (\mathrm{i}yc + s)e^{\mathrm{i}xy} - \mathrm{i}y + A(y^2 - 1)$. With $q_v = 2B^2/3$, $r_v = 2BD/3 - 4B^2/9$, $\lambda = 9/8$, and $\lambda_v = \lambda x^4 c^2/4$, we verify the global nonnegativity of

$$\mathcal{D}_x(y) := h(y)\{q_v + (r_v/2)(y^2 - 1) + \lambda_v h(y)\} - |V_x(y)|^2 \geq 0 \qquad (y \geq 0), \tag{B.1}$$

where $h(y) = (y - 1)^2(y + 1/2)$. Because $\mathcal{D}_x(y)$ vanishes to fourth order at $y = 1$, we factor out $(y - 1)^4$ explicitly. Using $E_n(\mathrm{i}v) = \sum_{m \geq 0} (\mathrm{i}v)^m/(m + n)! = ((n - 1)!)^{-1} \int_0^1 (1 - u)^{n-1} e^{\mathrm{i}vu}\, du$ and $z = y - 1$, we have

$$\frac{\mathcal{D}_x(y)}{(y - 1)^4} = \lambda_v(y + 1/2)^2 + r_v(y/2 + 5/4) - A^2 - \Re \sum_{n=1}^{4} c_n E_n(\mathrm{i}xz),$$

where $c_4 = -2x^4$, $c_3 = 2x^4 + 2\mathrm{i}x^3(1 + c^2)$, $c_2 = x^2(4Asc + 2c^2) - 2\mathrm{i}x^2\{A(1 + 2c^2) - sc\}$, and $c_1 = -2xAc^2 - 2\mathrm{i}xAsc$. Dividing further by x^4 removes the fourth-order vanishing at $x = 0$: with $S = \operatorname{sinc} x$, $u = \operatorname{sinc}(2x) = Sc$, $a = (1 + u)/2$, $R = \frac{2}{3}S^2(1 + 2u) - \frac{4}{9}S^4$, and $K_0 = -uS^2 - 4x^2[\Re E_3(2\mathrm{i}x)]^2$, the quotient $\mathcal{H}(x, y) = \mathcal{D}_x(y)/[x^4(y - 1)^4]$ is

$$\begin{aligned}\mathcal{H}(x, y) = {} & \frac{\lambda}{4}c^2(y + 1/2)^2 + R(y/2 + 5/4) + K_0 - 2au(z + 2)\Re E_2(\mathrm{i}xz) \\ & - 2\{ac^2z^2 + [a(1 + 2c^2) - u]z + 1\}\Re E_3(\mathrm{i}xz) + 2\{c^2z^2 + (1 + c^2)z + 1\}\Re E_4(\mathrm{i}xz),\end{aligned} \tag{B.2}$$

with $\mathcal{H}(0, y) = (4y + 1)/144 + (\lambda - 1)(y + 1/2)^2/4 > 0$.

The certificate `certificates/scalar/energy.json` covers $(x, y) \in [0, 1/2] \times [0, 48]$ using (B.2) and $(x, y) \in [1/2, 1] \times [0, 16]$ using $\mathcal{D}_x(y)/(y - 1)^4$. On the unbounded spatial tails, $|U_x(y)| \leq P(y) := y^3 + \frac{20}{21}y^2 + \frac{8}{7}y + \frac{4}{21}$ for $0 \leq x \leq 1/2$, and $\lambda h(y)^2 - P(y)^2$ has strictly positive expansion coefficients in $z = y - 48 \geq 0$; similarly, for $1/2 \leq x \leq 1$ and $y = 16 + z \geq 16$, $\mathcal{D}_x(y) \geq \frac{1}{100}(y - 1)^4(y + 1/2)^2 - (3y^2/4 + 2y + 1)^2$ has strictly positive coefficients in $z \geq 0$. Dividing (B.1) by $h(Y)$ and applying Cauchy–Schwarz to $\mathbb{E}T_x(W) = \mathbb{E}U_x(W)$ yields (7.4).

References

[1] C.-G. Esseen. A moment inequality with an application to the central limit theorem. *Skandinavisk Aktuarietidskrift*, 39:160–170, 1956. doi: 10.1080/03461238.1956.10414946.

[2] Larry Goldstein. Bounds on the constant in the mean central limit theorem. *The Annals of Probability*, 38(4):1672–1689, 2010. doi: 10.1214/10-AOP527. URL https://arxiv.org/abs/0906.5145v2.

[3] Larry Goldstein and Gesine Reinert. Stein's method and the zero bias transformation with application to simple random sampling. *The Annals of Applied Probability*, 7(4):935–952, 1997. doi: 10.1214/aoap/1043862419. URL https://arxiv.org/abs/math/0510619.

[4] Fredrik Johansson. Arb: Efficient arbitrary-precision midpoint-radius interval arithmetic. *IEEE Transactions on Computers*, 66(8):1281–1292, 2017. doi: 10.1109/TC.2017.2690633. URL https://arxiv.org/abs/1611.02831.

[5] Fredrik Johansson. Numerical integration in arbitrary-precision ball arithmetic. In *Mathematical Software – ICMS 2018*, volume 10931 of *Lecture Notes in Computer Science*, pages 255–263. Springer, 2018. doi: 10.1007/978-3-319-96418-8_30. URL https://arxiv.org/abs/1802.07942.

[6] Lutz Mattner and Irina Shevtsova. An optimal Berry–Esseen type theorem for integrals of smooth functions. *ALEA, Latin American Journal of Probability and Mathematical Statistics*, 16:487–530, 2019. URL https://alea.impa.br/articles/v16/16-19.pdf.

[7] NIST Digital Library of Mathematical Functions. Section 4.22: Infinite products and partial fractions, n.d. URL https://dlmf.nist.gov/4.22.E3. Equation 4.22.3. Accessed 7 September 2026.

[8] Iosif Pinelis. More on the nonuniform Berry–Esseen bound, 2013. URL https://arxiv.org/abs/1302.0516v3. Version 3.

[9] H. Prawitz. Limits for a distribution, if the characteristic function is given in a finite domain. *Scandinavian Actuarial Journal*, 1972(2):138–154, 1972. doi: 10.1080/03461238.1972.10404645.

[10] H. Prawitz. Weitere ungleichungen für den absoluten betrag einer charakteristischen function. *Scandinavian Actuarial Journal*, 1975(1):21–28, 1975. doi: 10.1080/03461238.1975.10405075.

[11] I. G. Shevtsova. On the absolute constants in the Berry–Esseen inequality and its structural and nonuniform improvements. *Informatics and its Applications*, 7(1):124–125, 2013. URL https://www.mathnet.ru/eng/ia252. In Russian.

[12] Irina Shevtsova. Moment-type estimates with asymptotically optimal structure for the accuracy of the normal approximation. *Annales Mathematicae et Informaticae*, 39:241–307, 2012. URL https://publikacio.uni-eszterhazy.hu/3231/1/AMI_39_from241to307.pdf.

[13] Ilya S. Tyurin. New estimates of the convergence rate in the Lyapunov theorem, 2009. URL https://arxiv.org/abs/0912.0726v1.